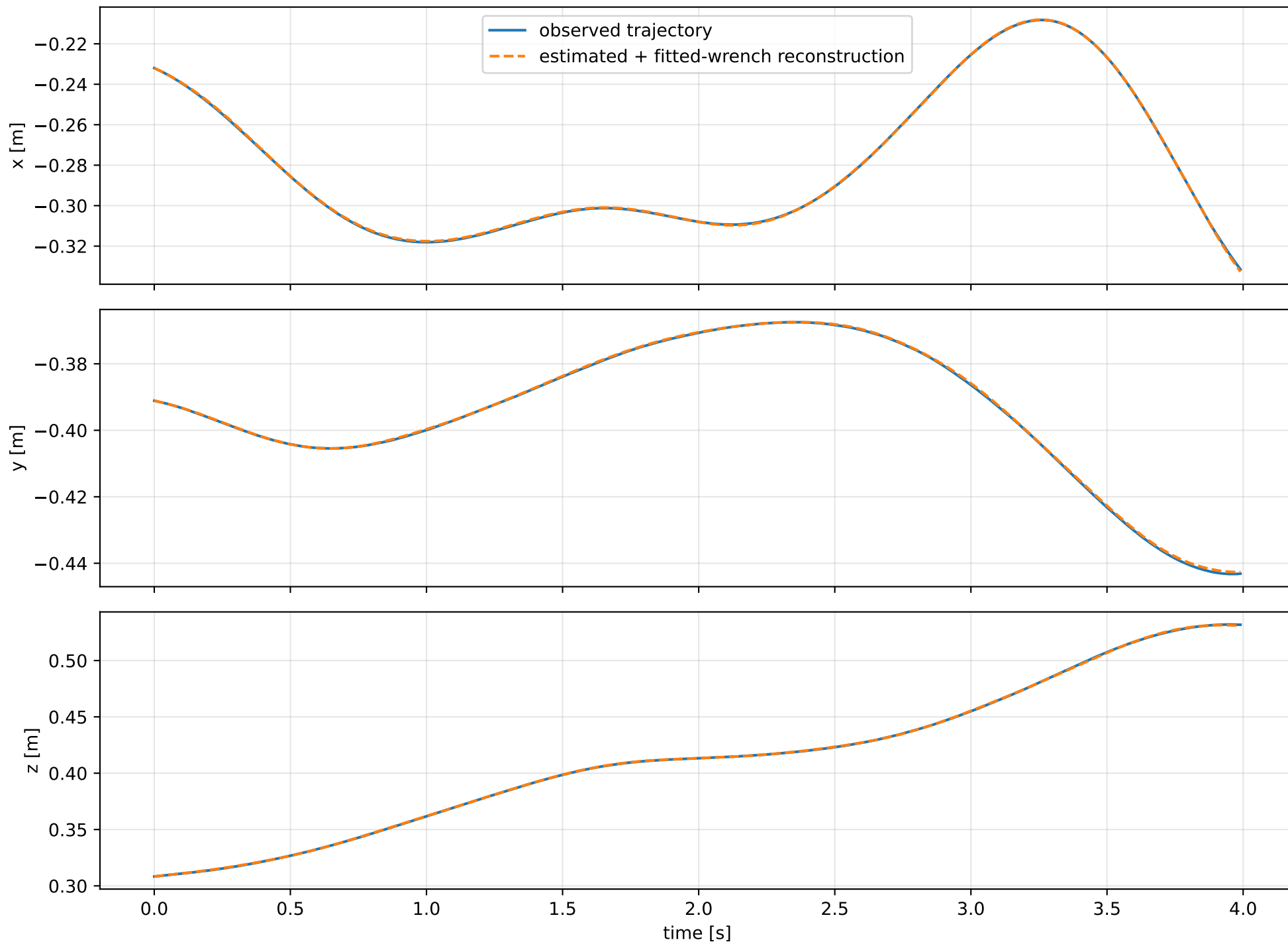
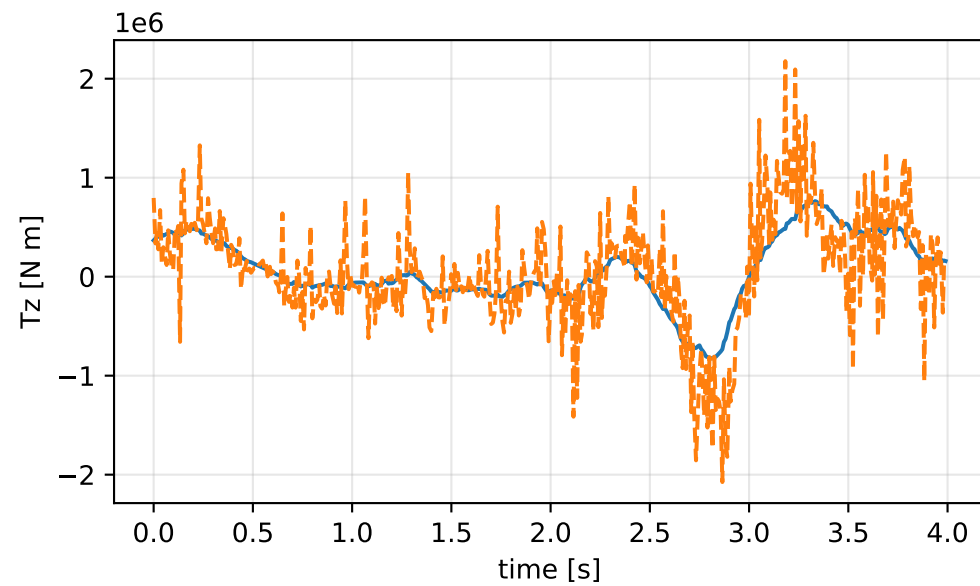
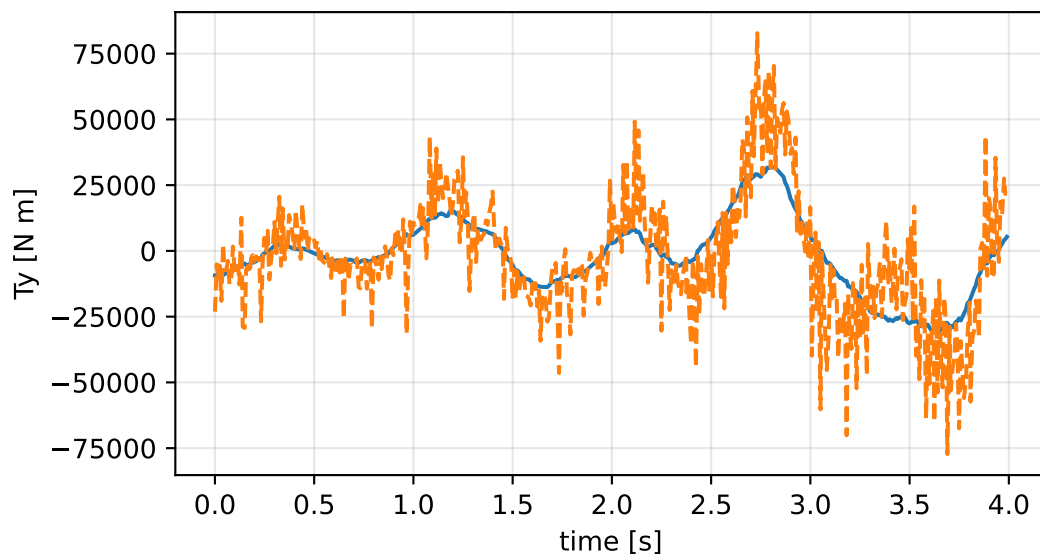
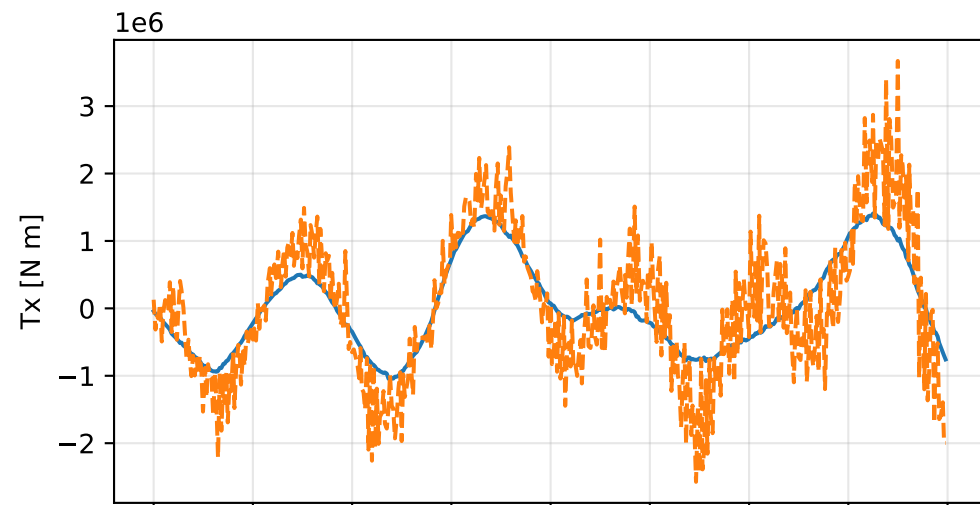
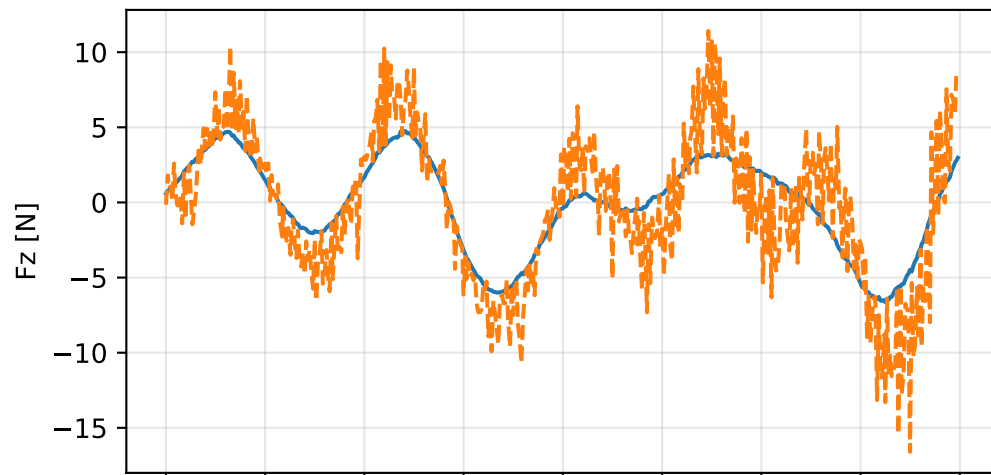
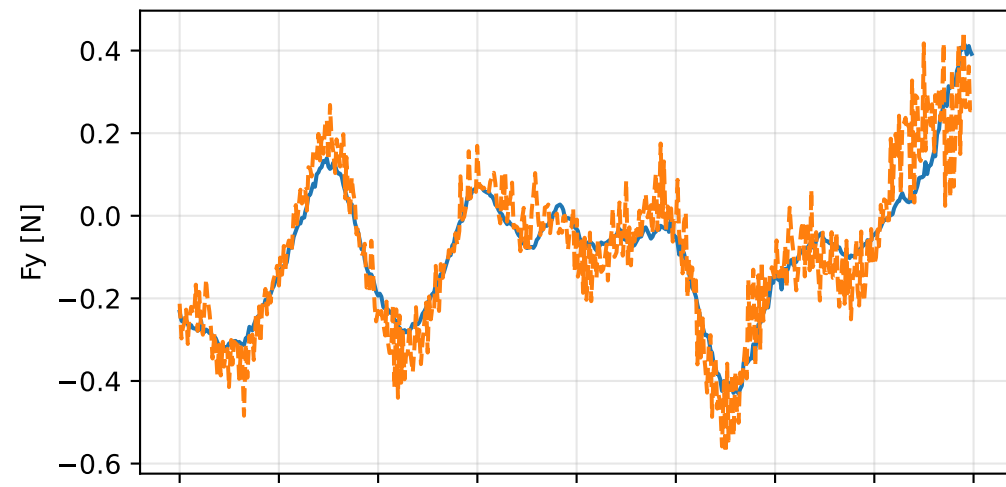
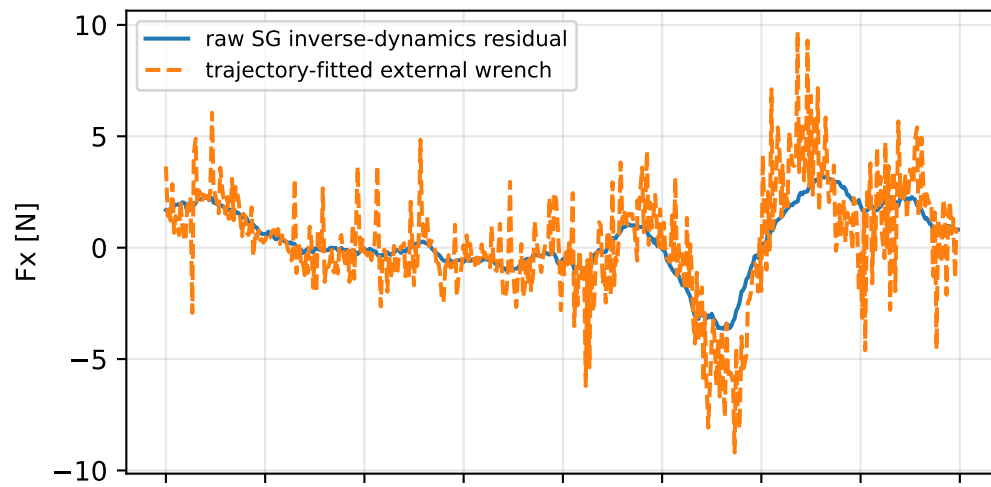


sweep_sg_window_covariance__003: trajectory

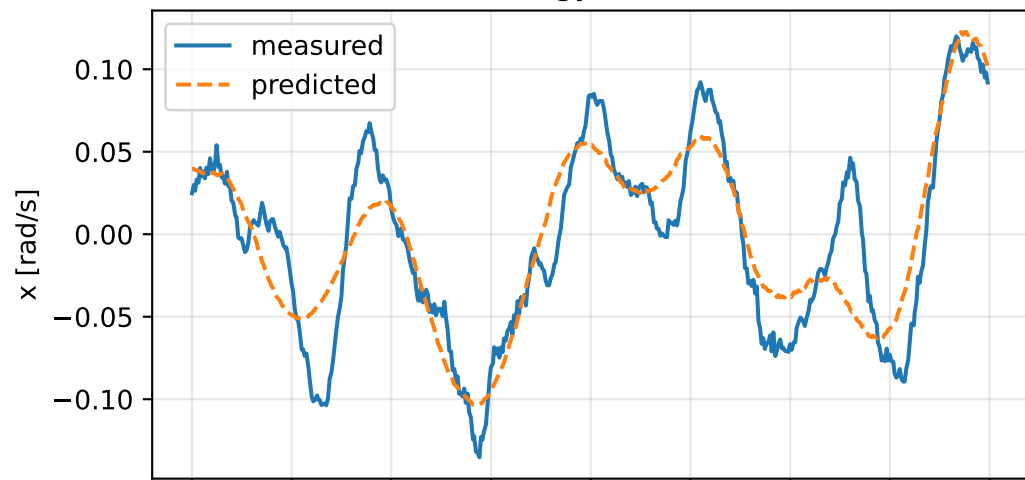


sweep_sg_window_covariance_003: residual wrench

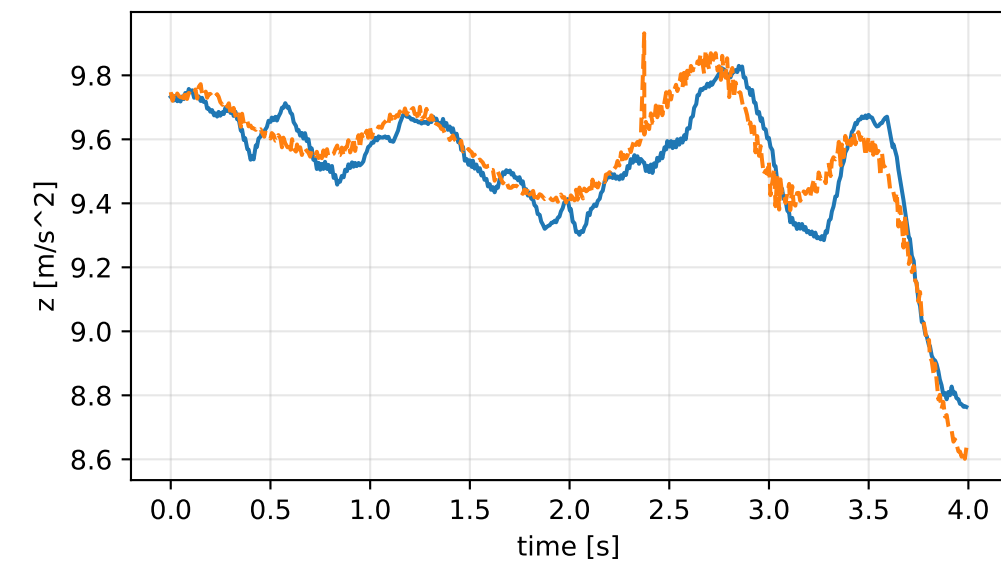
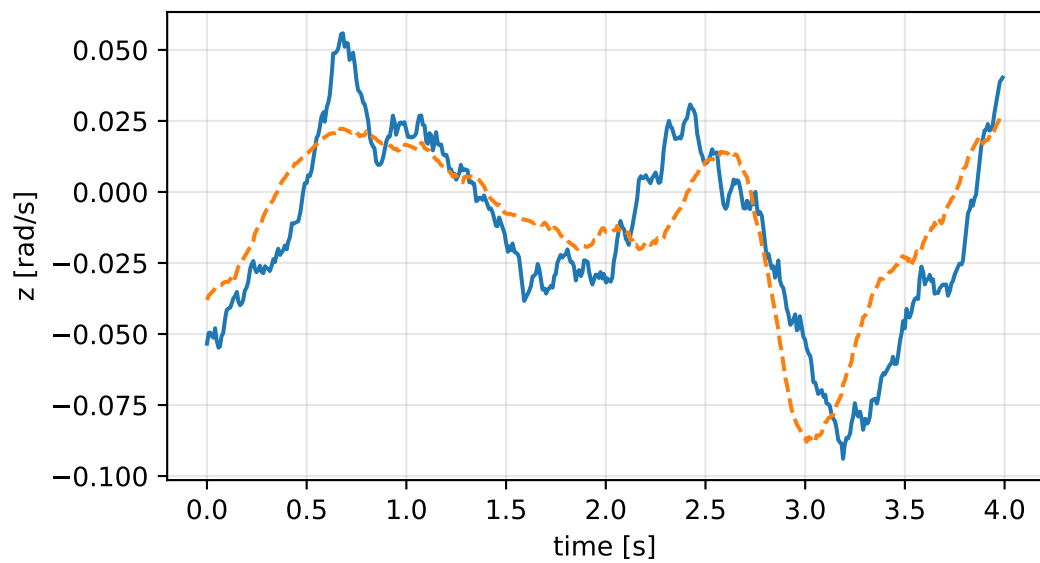
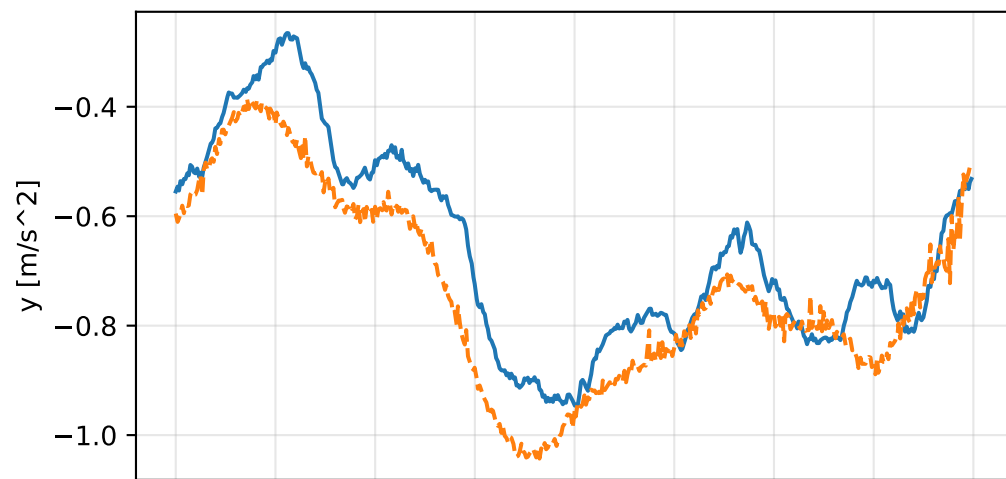
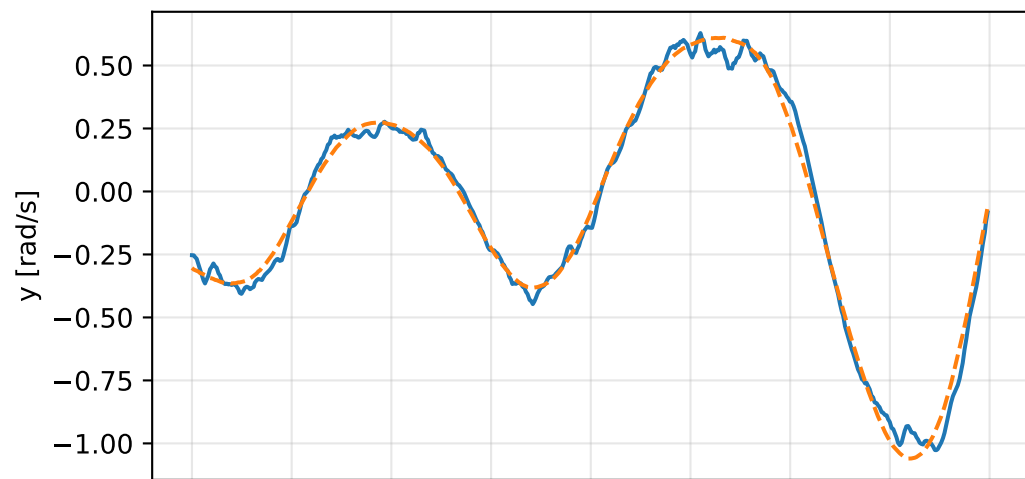
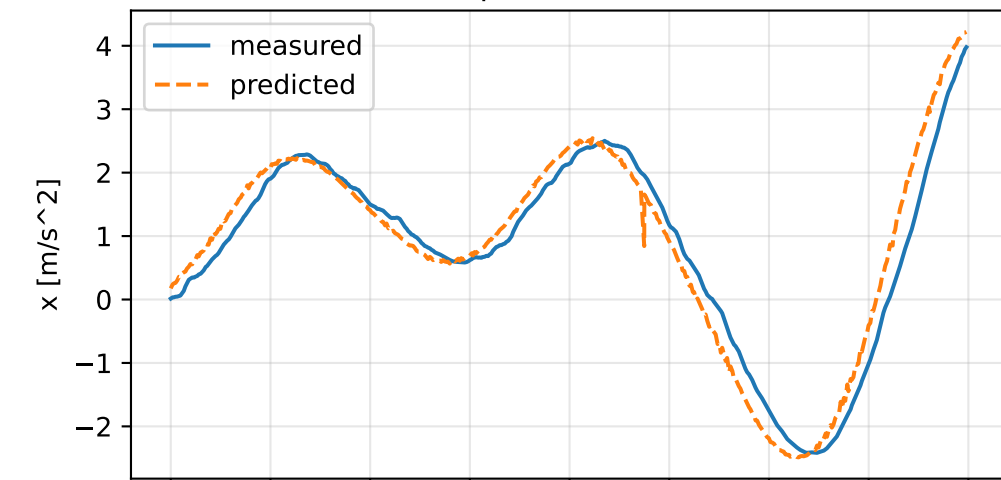


sweep_sg_window_covariance_003: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__003: nominal -> estimated parameters

Case: sweep_sg_window_covariance__003
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 1.53207428 delta -0.819523313

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [3.8543191e+06 -4.8919349e+04 -1.3209132e+03] d=[3.8543191e+06 -4.8919349e+04 -1.3209132e+03]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [-48919.3493579 3429.4121704 -103995.6012369] d=[-48919.3493572 3429.4121704 -103995.6012369]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-1.3209132e+03 -1.0399560e+05 3.8521325e+06] d=[-1.3209133e+03 -1.0399560e+05 3.8521325e+06]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.1524665 -11.3365345 -0.290202] d=[-0.1504418 -11.3365345 -0.290202]

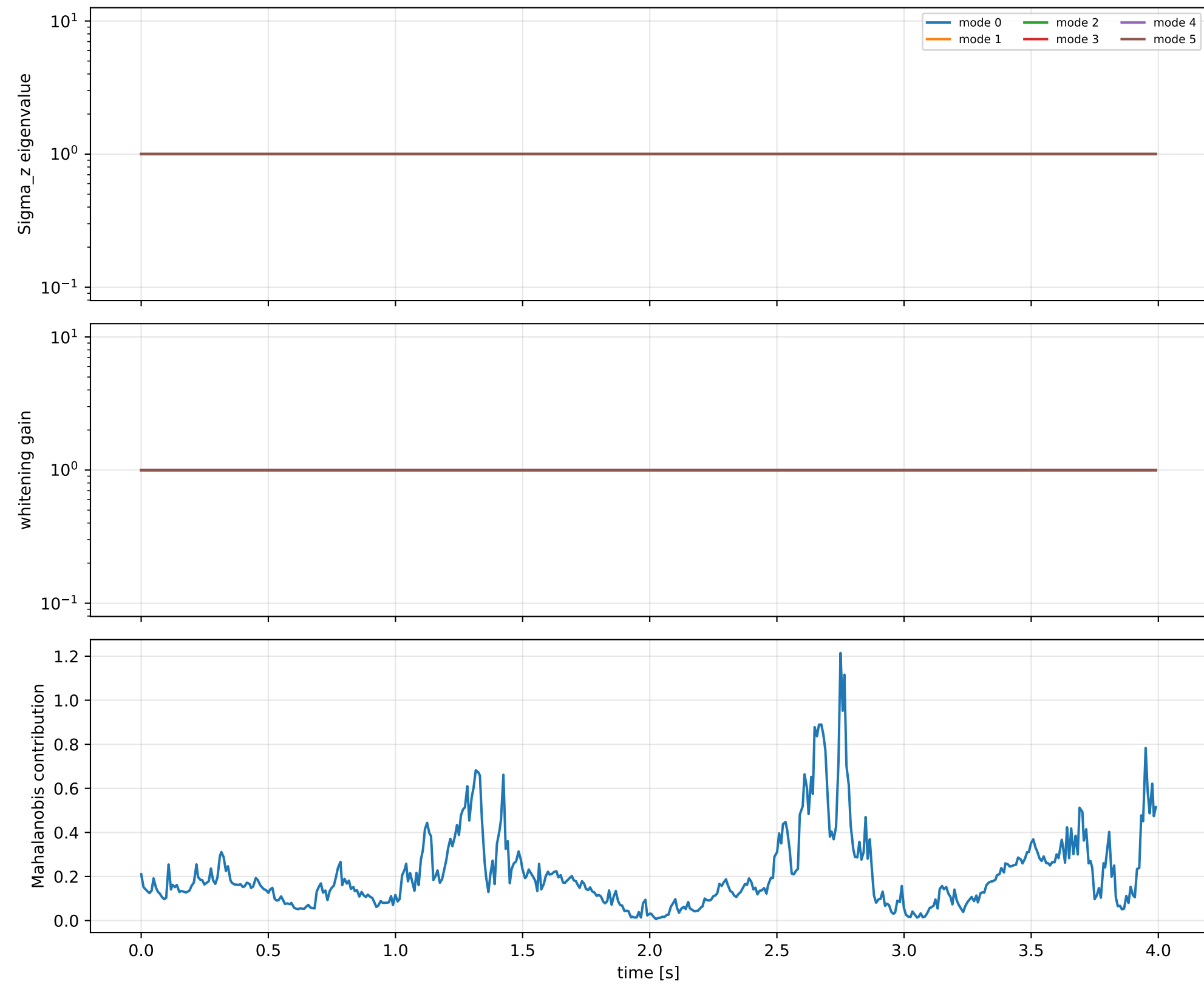
rotor force effectiveness

[1. 1. 1. 1.] -> [1.1170913e+00 1.0828746e+00 2.5554114e-05 2.6381278e-02] d=[0.1170913 0.0828746 -0.9999744 -0.9736187]

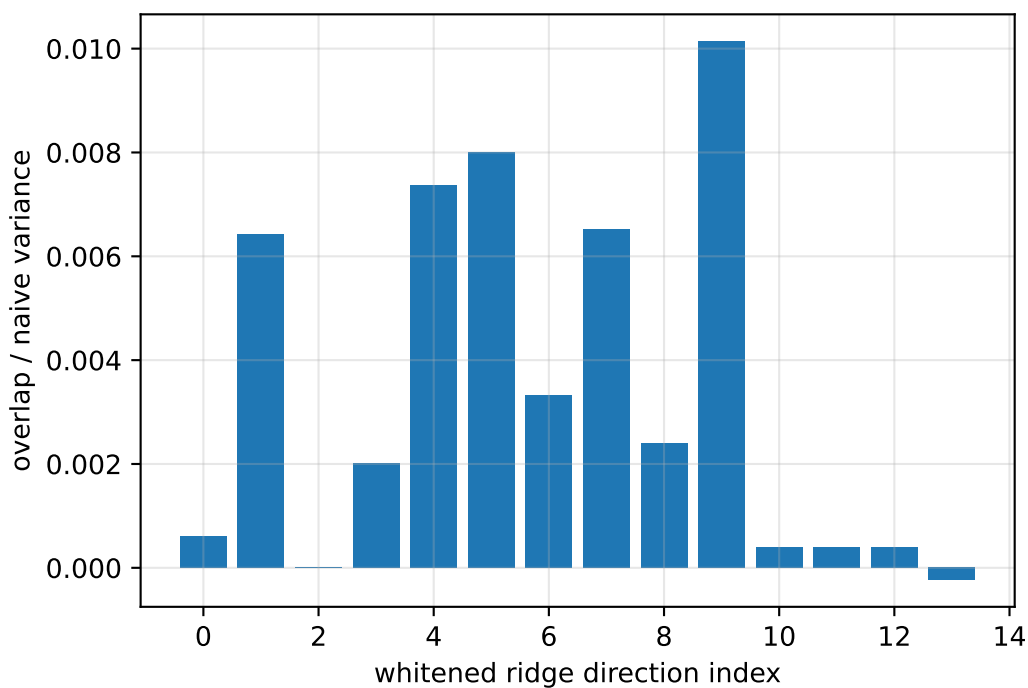
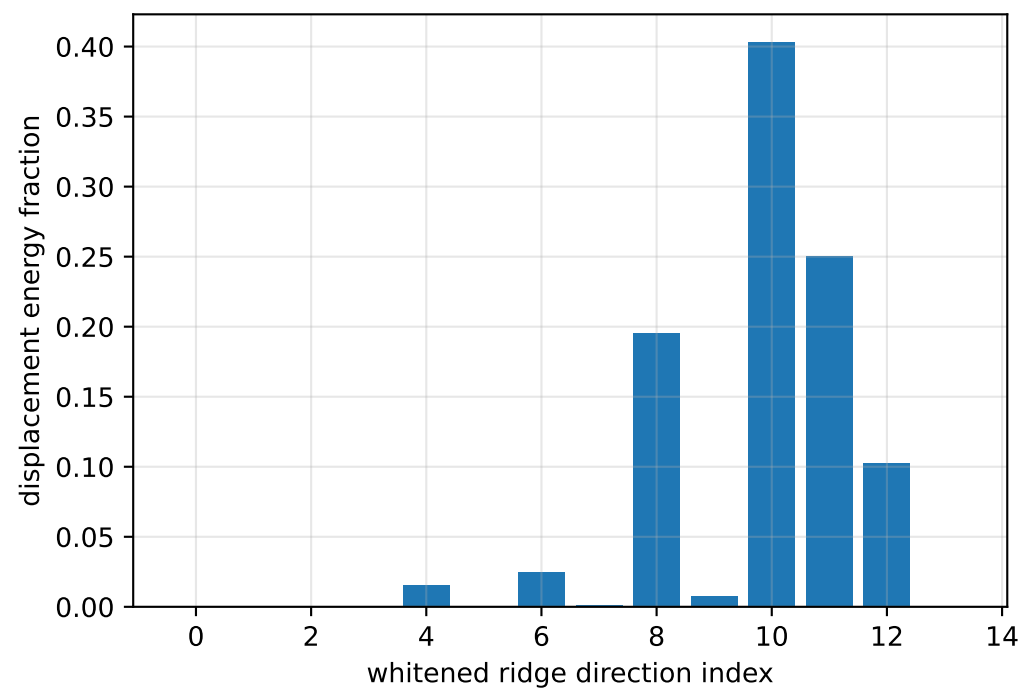
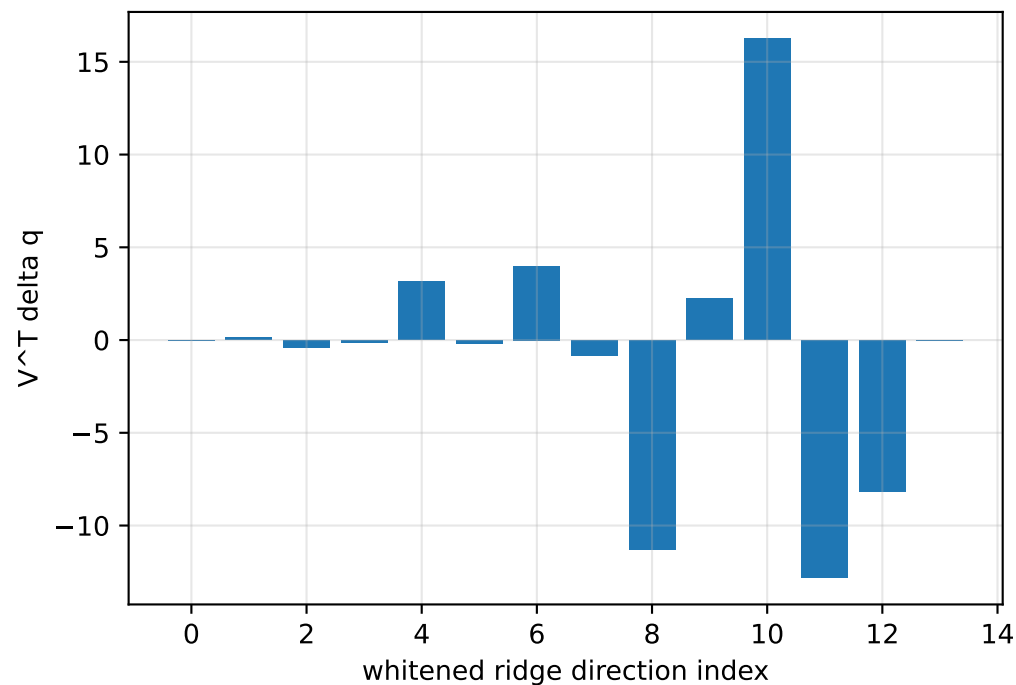
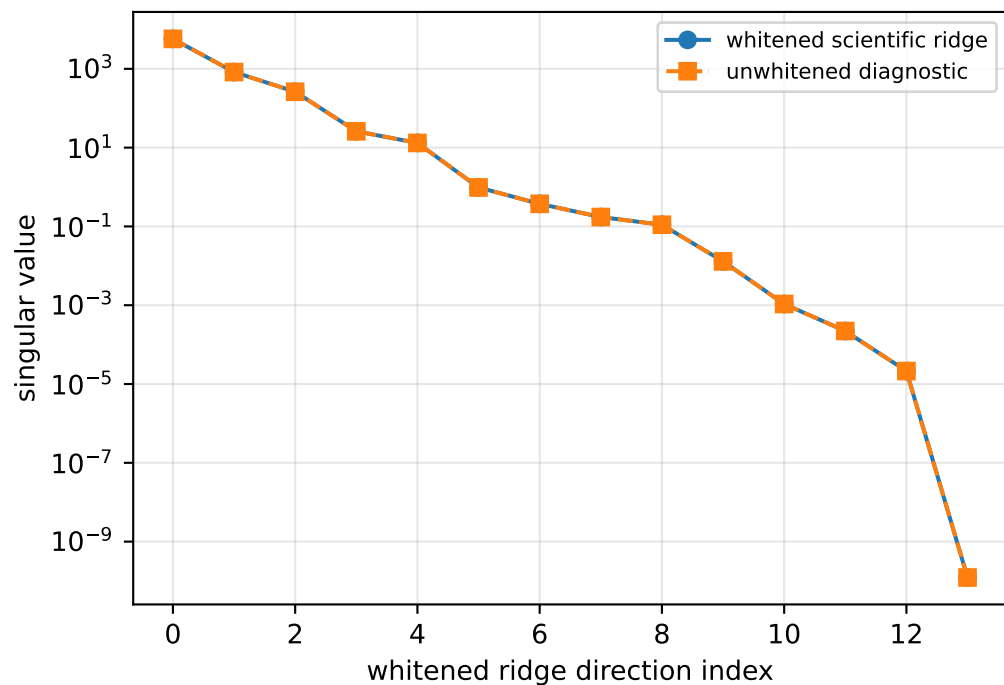
rotor lag [s] 0.2

gimbal lag [s] 0.0148715973

sweep_sg_window_covariance__003: covariance weighting

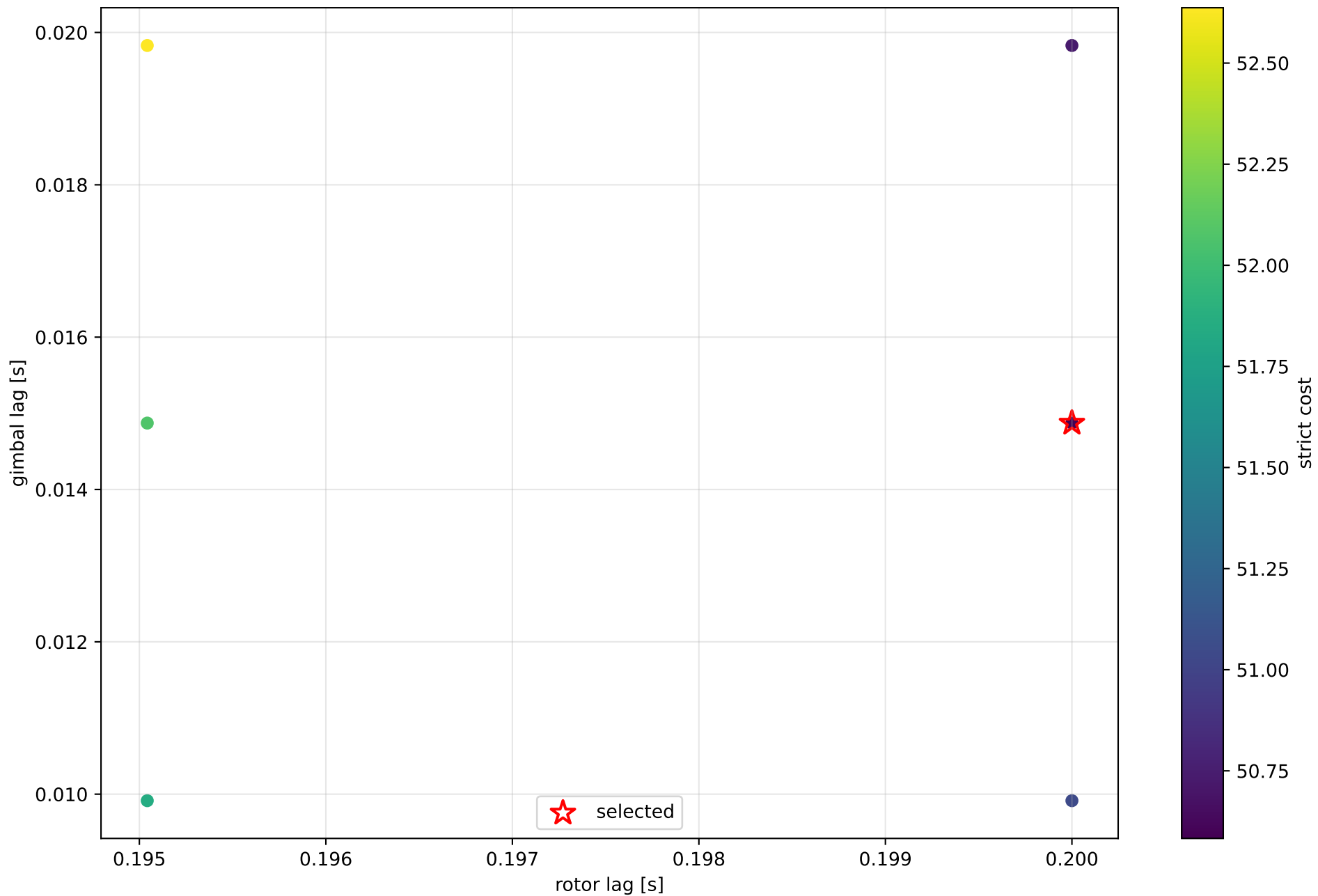


sweep_sg_window_covariance_003: parameter ridge



exact scale gauge: $\|v_{\text{scale}}\|=1.51637\text{e-}07$, rank=13, nullity=1

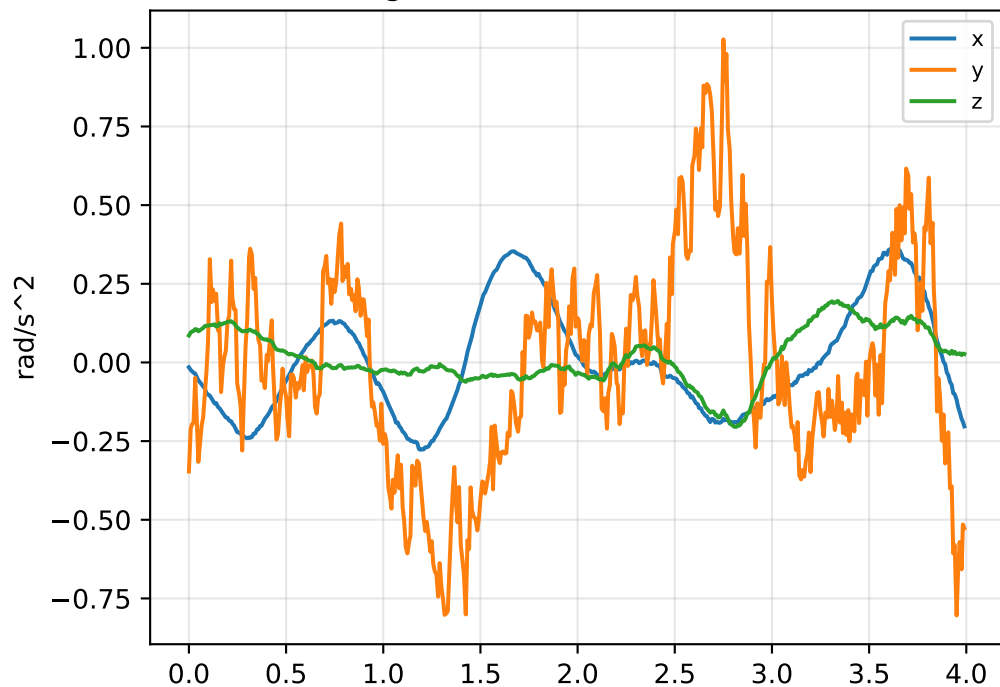
sweep_sg_window_covariance__003: strict lag candidates



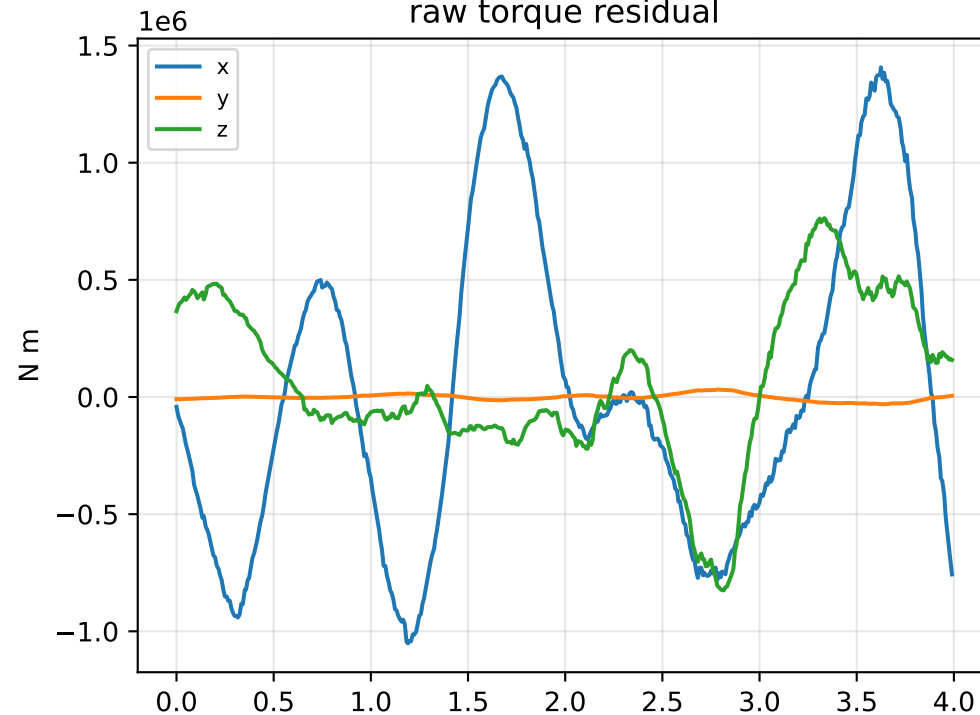
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep_sg_window_covariance_003: acceleration/wrench closure

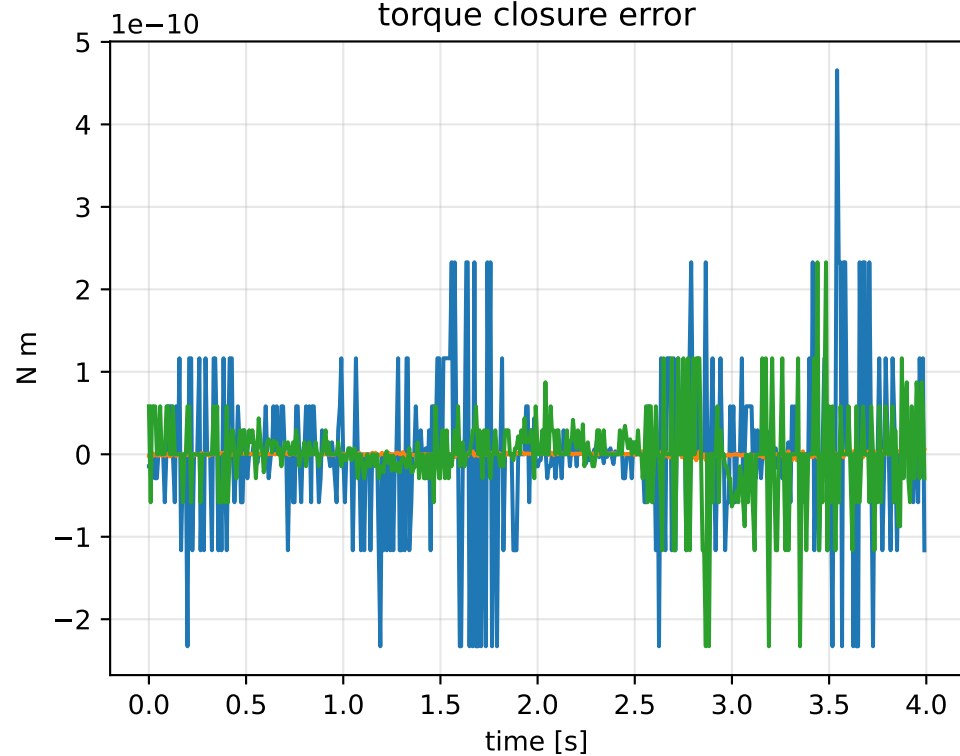
angular-acceleration residual



raw torque residual



torque closure error



principal moments [kg m²]
 nominal: [0.064953 0.065 0.128992]
 estimated: [5.997817e-02 3.854940e+06 3.854941e+06]
 estimated / nominal: [9.234138e-01 5.930671e+07 2.988509e+07]

force closure max/rms: 4.46865e-15 / 1.01901e-15
 torque closure max/rms: 4.65661e-10 / 5.759e-11